# EE3032 Introduction to VLSI Design—Final

Fall 2008

Reconstructed from the complete three-page scanned original, printed pages 1-1 through 1-3. Question weights total **115%**. The original Fall 2008 footer controls the year; handwritten annotations are not transcribed as prompt text. All source circuit polarities, full function overbar and sizing labels are retained in editable TikZ. The reconstruction preserves the printed mathematical premises.

**1. (30%)** Answer the following problems.

(a) List two methods for reducing the power bounce and ground bounce in the IO pads. (4%)

(b) List major steps of the CMOS full-custom design flow. (6%)

(c) List two methods for reducing the switching power of a CMOS circuit. (4%)

(d) Explain the energy consumption of an inverter when the inverter performs a switching. (4%)

(e) List two major problems of dynamic logic. (4%)

(f) List two limitations of CMOS domino logic. (4%)

(g) What are core-limited design and pad-limited design? (4%)

**2. (15%)** Sketch the Boolean function $f = \overline{A(B+C)+DE}$ using each of the following circuit techniques. (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%).

**3. (10%)** Explain the operation principle of the circuit shown in Figure 1.

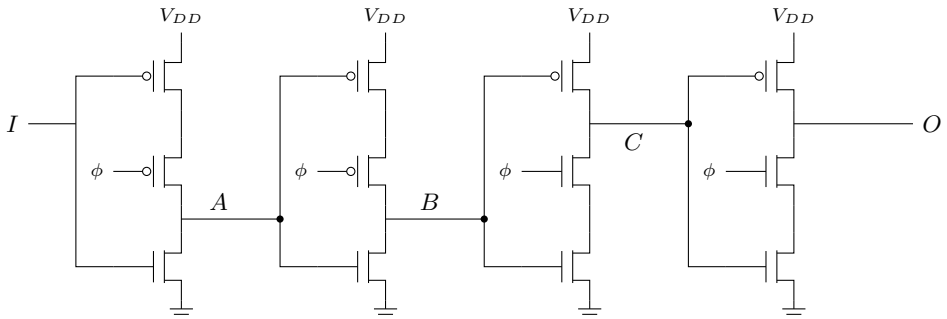



Figure 1: A dynamic CMOS register circuit. All four three-device stages and the two first-stage clock-controlled pMOS bubbles are retained.

**4. (15%)** The circuit shown in Figure 2 results from the implementation of a high-speed one-bit full adder. Complete the following problems.

(a) What type of CMOS logic style is used to implement this circuit? (5%)

(b) Suppose that the drain capacitance and gate capacitance of the PMOS and NMOS are $0.2\,fF$ and $1\,fF$, respectively. The leakage current of the output nodes $\bar{C}_{n+1}$ and $\bar{S}_n$ is $1\,nA$. Also, $V_{dd} = 1.8\,V$ and the switching threshold voltage of the inverter is $1\,V$. Determine the lowest speed at which the circuit should be operated. (10%)

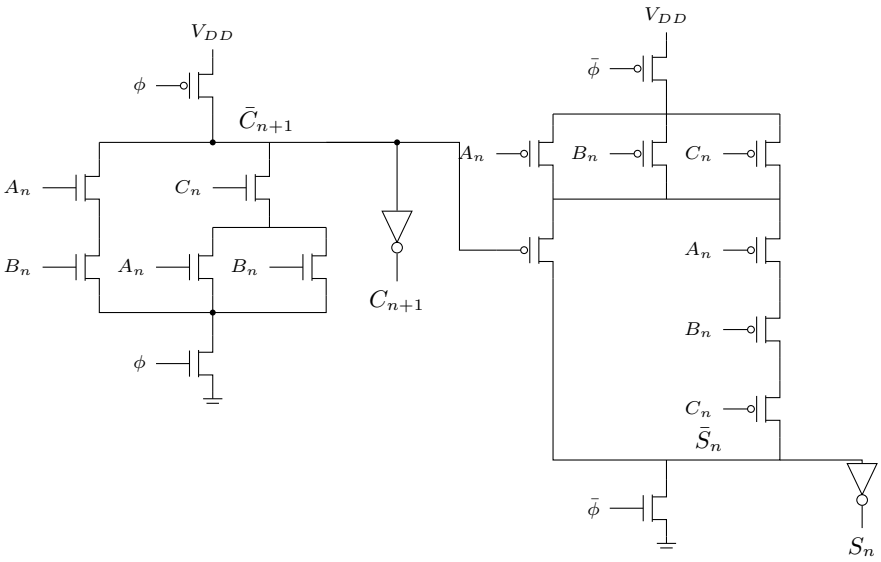



Figure 2: An adder circuit. The printed static output inverters are retained.

**5. (10%)** Figure 3 shows a sequential circuit (pipelined circuit). Assume that your boss asks you to design the circuit such that the circuit can be operated at $100\,MHz$. Also, assume that the clock-to-Q propagation delay, the setup time, and the hold time of registers R1 and R2 are $1\,ns$, $1.5\,ns$, and $1\,ns$, respectively. How fast should the combinational circuit be designed?

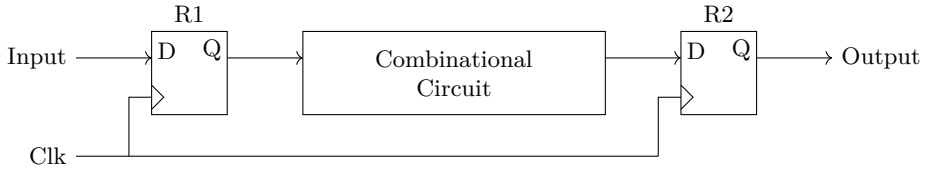



Figure 3: A pipelined circuit.

**6. (10%)** Figure 4 shows the Schmitt CMOS circuit and its voltage transfer curve. Assume that the Schmitt circuit is designed as $V_{T+} = 0.7V_{dd}$ and $V_{T-} = 0.3V_{dd}$. Also, the size of N1 is $(W/L)_{n1} = 2\,\mu m/0.5\,\mu m$ and the size of P1 is $(W/L)_{p1} = 3\,\mu m/0.5\,\mu m$. What sizes of N3 and P3 should be designed?

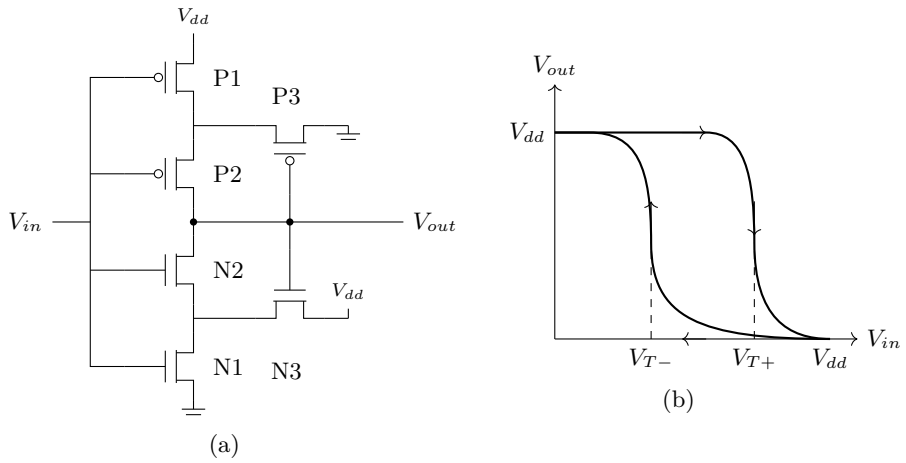



Figure 4: (a) A CMOS Schmitt circuit. (b) Its voltage transfer curve.

**7. (10%)** Figure 5 shows the symbolic layouts. Draw the schematic diagrams of the layouts.

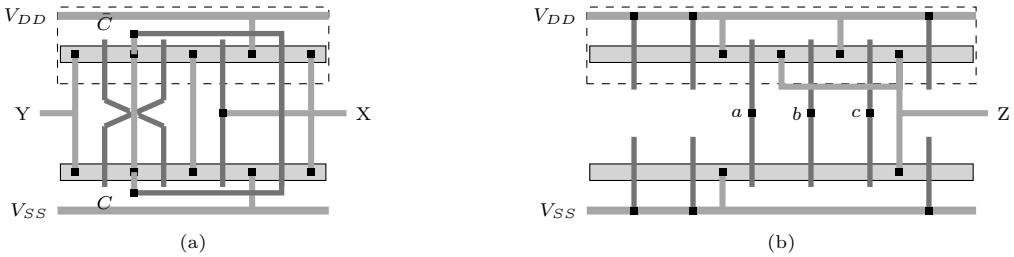



Figure 5: Symbolic layouts, with normalized spacing and source net connectivity. Grey horizontal bands: diffusion; light routes: Metal 1; dark narrow routes: poly; black squares: contacts; dashed boundary: n-well. The opposite clock routes in (a) remain separate.

**8. (15%)** Complete the following problems.

(a) Figure 6(a) shows a positive-active static latch design, and transistor sizes of the gate in the upstream of the latch are shown in the figure. Assume that the threshold of the two inverters is $V_{dd}/2$ and that the parasitic resistance/capacitance of the transmission gate can be ignored. Also, $\mu_n = 2\mu_p$. Determine the transistor sizes of inverter 2 such that the latch can work correctly. (10%)

(b) Use the two latches shown in Figure 6(c) to realize a negative edge-triggered register and draw the corresponding timing diagram. (5%)

*Source panel note:* The two mux/buffer latch drawings are visibly labelled (b); no separate panel (c) is drawn. The text's (c) reference is preserved above as a panel-label typo. The two drawn latches provide the complete data-selection premise.

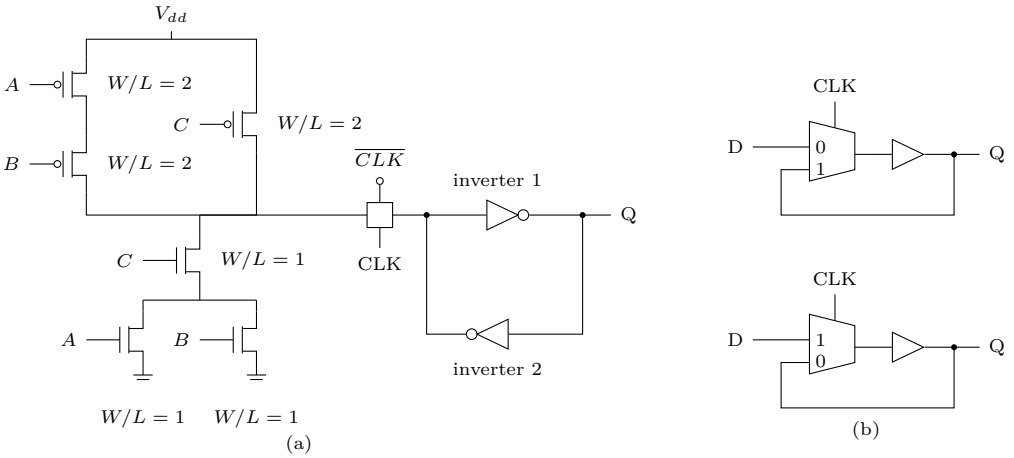



Figure 6: (a) Positive-active latch. (b) The two printed complementary-level latch drawings. The original caption/text also refer to panel (c), which is a panel-label typo; no third drawing is present.